

11. Rating Political Candidates

A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	$d = (R - 1)$	d^2
1	12 <i>7.5</i>	75 <i>3</i>	<i>4.5</i>	<i>20.25</i>
2	7 <i>4</i>	165 <i>7</i>	<i>-3</i>	<i>9</i>
3	5 <i>3</i>	300 <i>0</i>	<i>-4</i>	<i>16</i>
4	19 <i>12</i>	15 <i>1</i>	<i>11</i>	<i>121</i>
5	17 <i>10</i>	180 <i>8</i>	<i>2</i>	<i>4</i>
6	12 <i>7.5</i>	240 <i>11</i>	<i>-3.5</i>	<i>12.25</i>
7	9 <i>6</i>	120 <i>4</i>	<i>2</i>	<i>4</i>
8	18 <i>11</i>	60 <i>2</i>	<i>4</i>	<i>16</i>
9	3 <i>1</i>	230 <i>10</i>	<i>-4</i>	<i>16</i>
10	8 <i>5</i>	200 <i>9</i>	<i>4</i>	<i>16</i>
11	15 <i>9</i>	130 <i>5.5</i>	<i>3.5</i>	<i>12.25</i>
12	4 <i>2</i>	130 <i>5.5</i>	<i>-3.5</i>	<i>12.25</i>
<i>$\Sigma d^2 = 454$</i>				

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

$$a. \quad r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(454)}{12(144 - 1)} = 1 - \frac{2724}{12 \times 143} = -0.5874$$

$$b. \quad H_0 = \rho_s \geq 0$$

$$H_a = \rho_s < 0$$

$$n = 12, \quad \alpha = 0.05, \quad \text{臨界值} = 0.497$$

$$RR: \quad r \leq R_s: -0.497$$

$$\therefore -0.587 < -0.497. \quad GRR$$

Reject H_0

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)	$d = x - y$	d^2
1	9	33.15	6	36
2	13	33.33	7	49
3	5	33.50	0.5	0.25
4	7	33.55	2	4
5	12	33.73	3	9
6	6	33.86	-1	1
7	4	33.90	-5	25
8	5	34.15	-4.5	20.25
9	3	34.90	-7.0	49
			$\Sigma d^2 = 208.5$	

- a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

$$a. r_s = 1 - \frac{6(208.5)}{9(9^2 - 1)} = 1 - \frac{1251}{9 \times 80} = -0.7375$$

$$b. H_0: \rho_s = 0$$

$$H_a: \rho_s \neq 0$$

$$n = 9, \alpha = 0.05, 0.683$$

$$r_s = -0.7375, \text{ERR}$$

$$\therefore 0.7375 > 0.683, \text{ERR}$$

Reject H_0 .